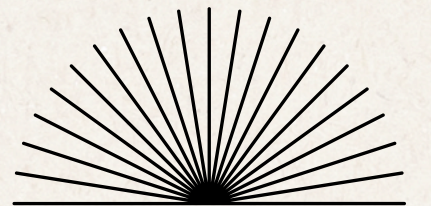


PREDICTING K-POP SONG POPULARITY:

**A Comparative Study of Regression,
Classification, and Company-Level Analysis**

GROUP 14

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Overview

In recent year, KPOP has already been a significant part of global music industry. We wanted to investigate what makes a KPOP song popular.

Therefore, we attempted to use Spotify Audio Features to predict whether a song is popular.

(generated by nano banana)



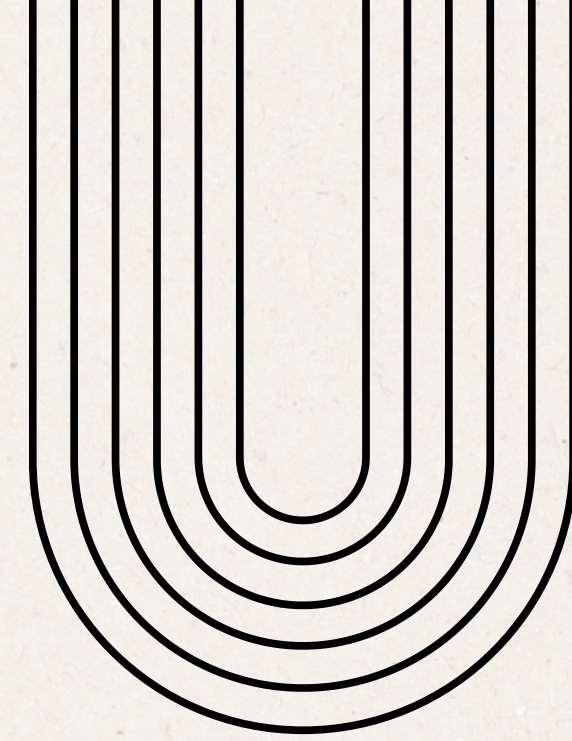
Methods

1 Data Collection

We use dataset from Kaggle, including melon-chart rankings, lyrics, audio features provided by Spotify, groups information.

2 Data Preprocessing

- We added the features “company,” “gender,” and “English lyrics ratio” to each data instance.
- For the “company” feature, we further processed the data by grouping smaller companies together while retaining major representative companies as separate categories.



3 Model Training

First part:

audio features, lyrics

Train regressors with audio features to predict average rank

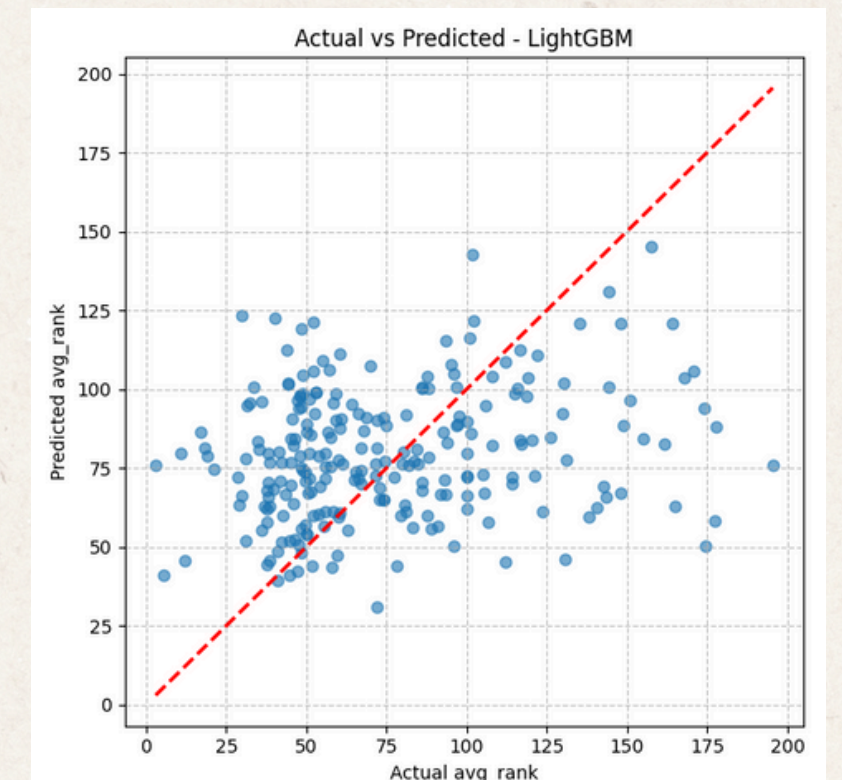
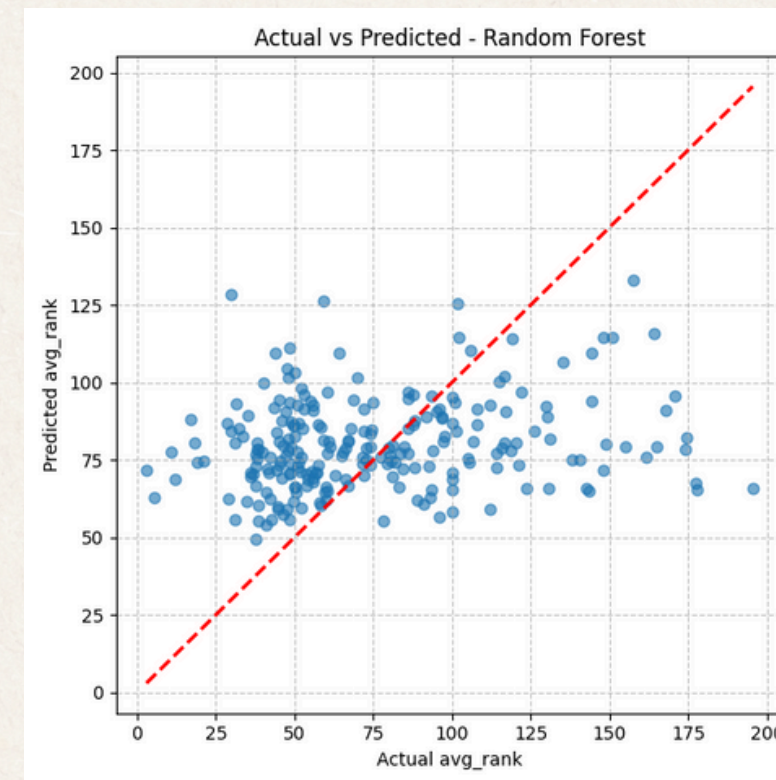
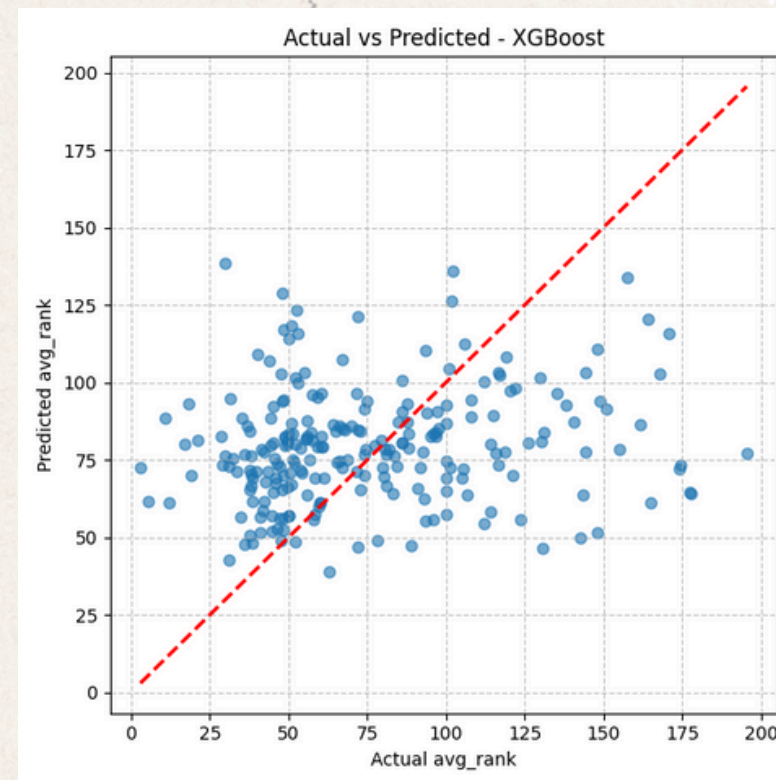
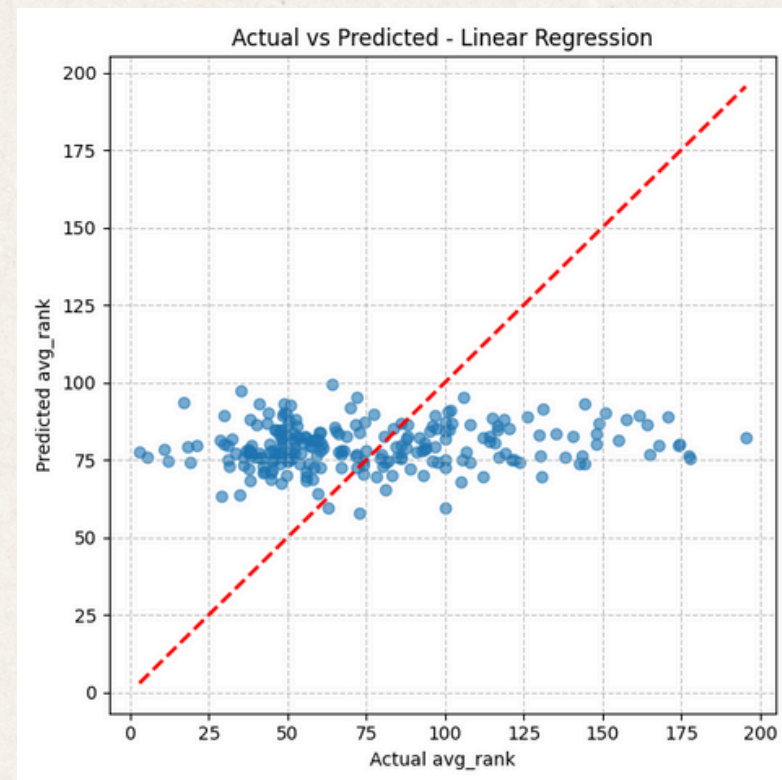
- Four regression models are used:

Linear Regression -> MSE: 1471.9802, MAE: 31.8449, R2: 0.0058

Random Forest -> MSE: 1479.6589, MAE: 30.5514, R2: 0.0006

XGBoost -> MSE: 1599.0760, MAE: 31.5336, R2: -0.0801

LightGBM -> MSE: 1561.3749, MAE: 31.1130, R2: -0.0546

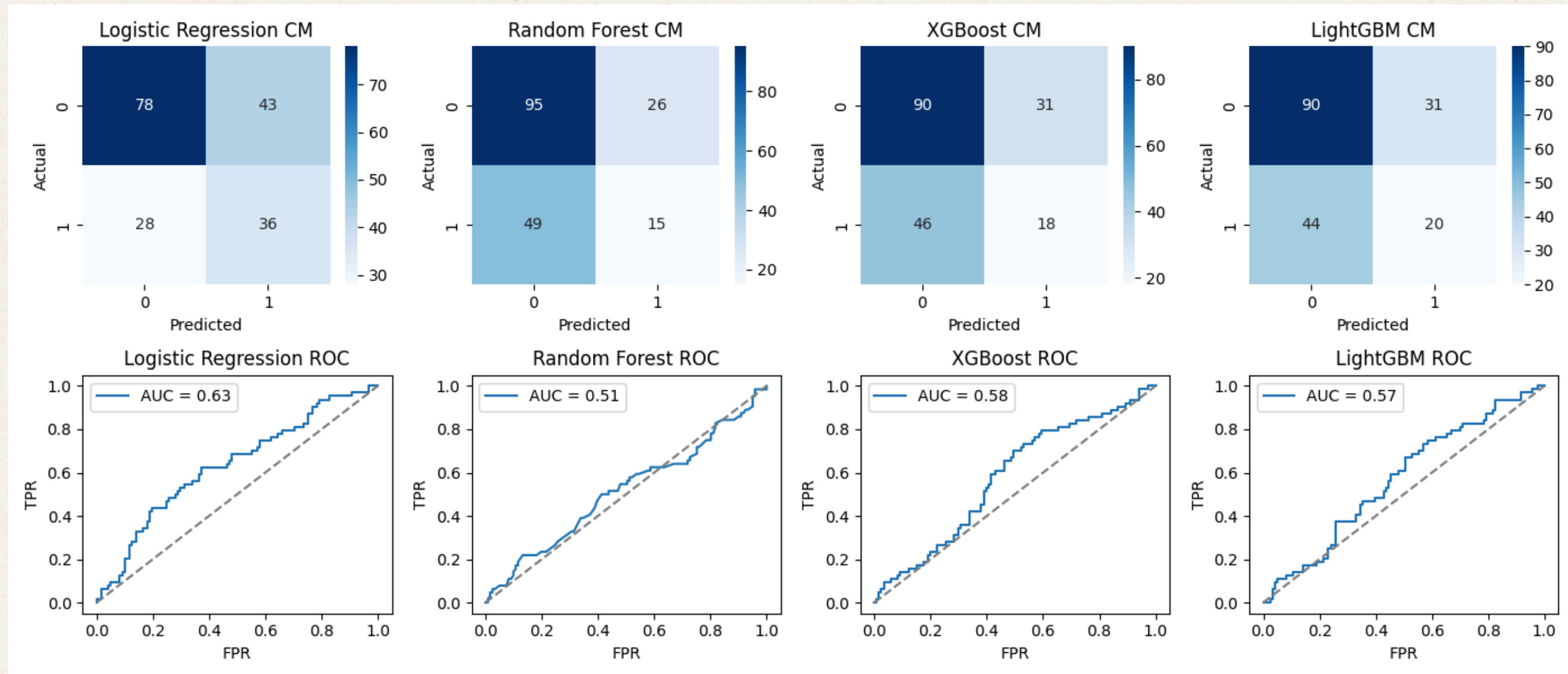


The predictive performance of these regressors is poor, regardless of the model used.

3 Model Training

Second part: add new features, train classifier

- Add company and gender feature
- Train a classifier instead of a regressor, to predict songs are in top 35 or not

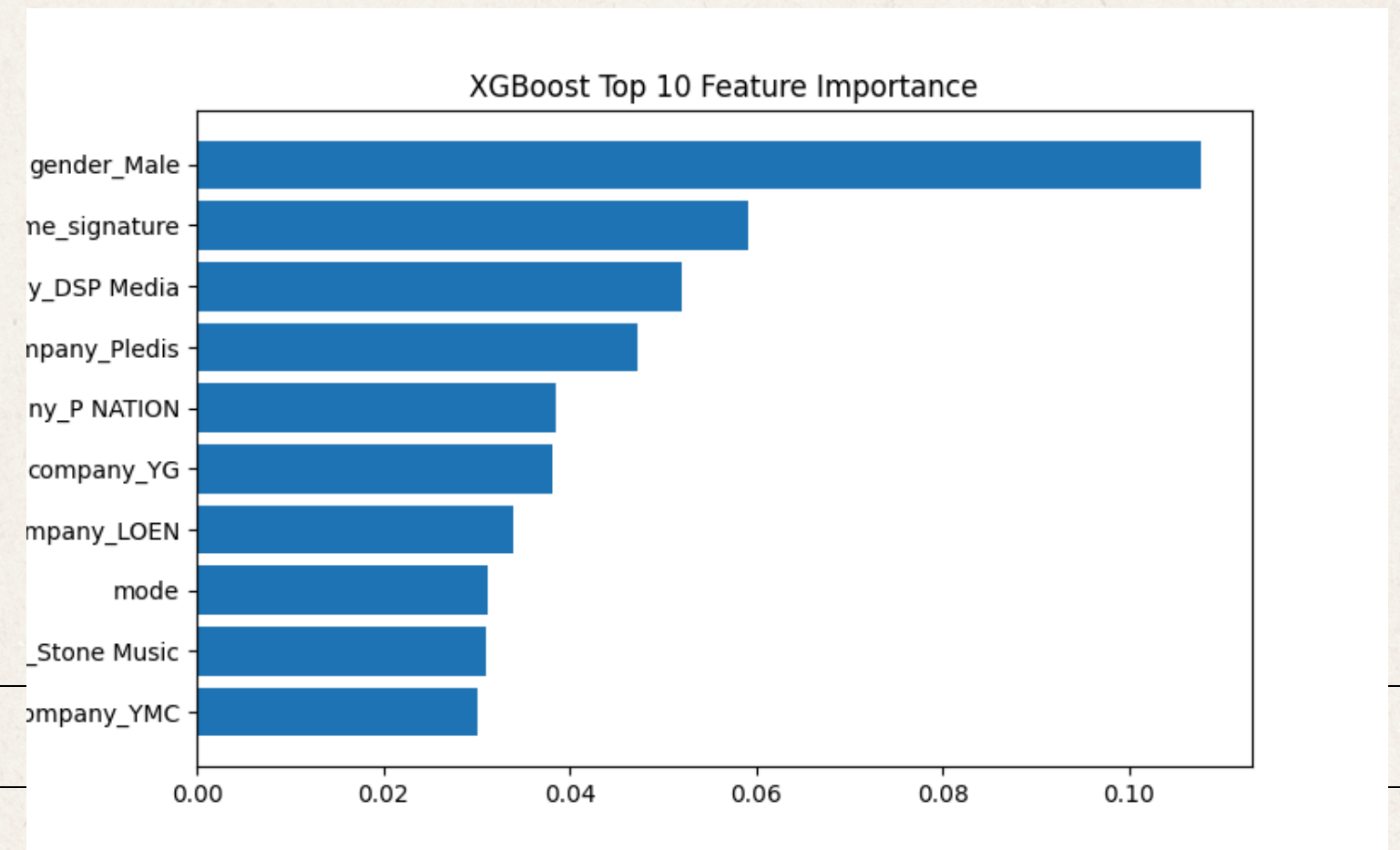
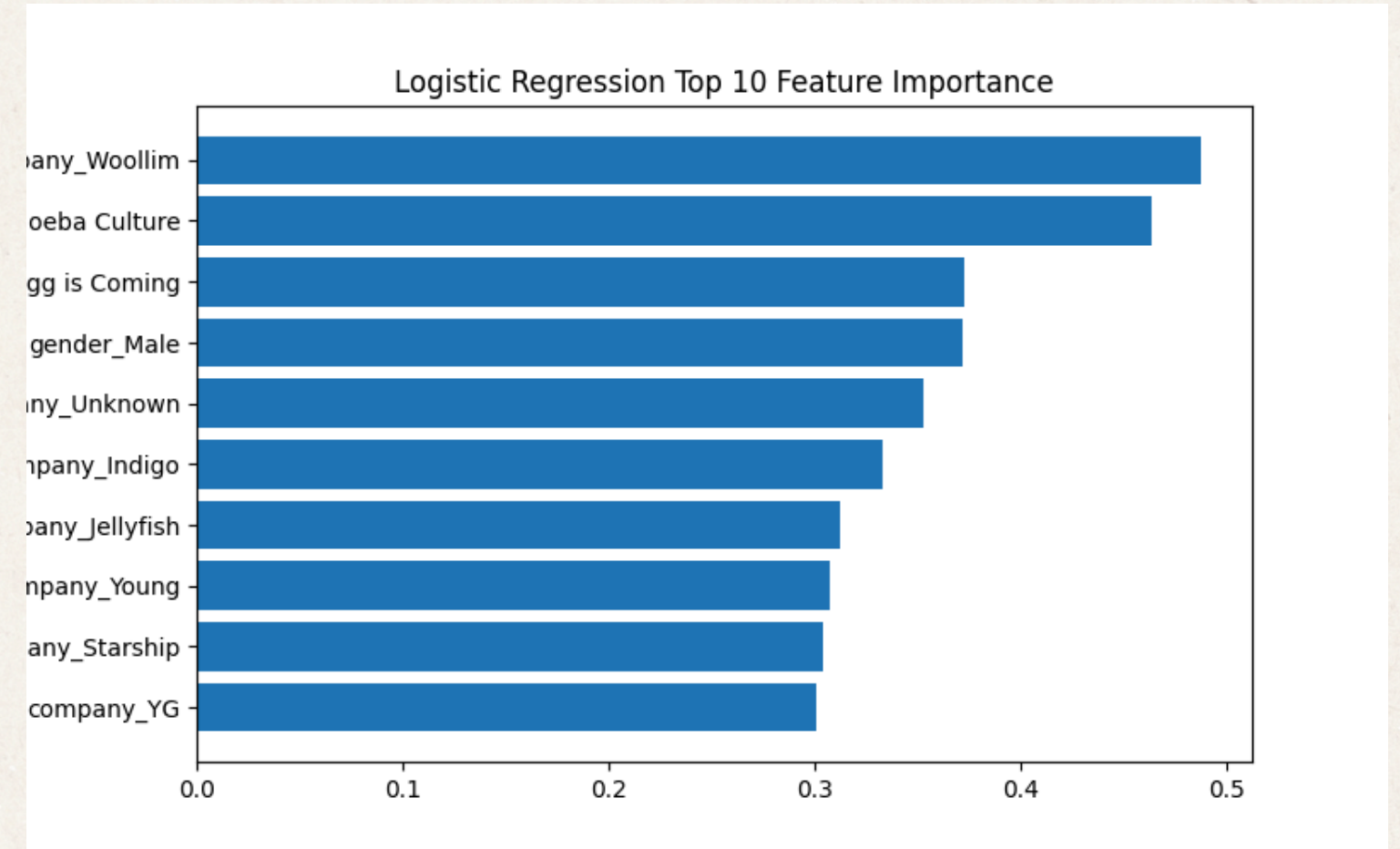


Discussion

Audio Features Are Ineffective & Company Greatly Impacts Ranking

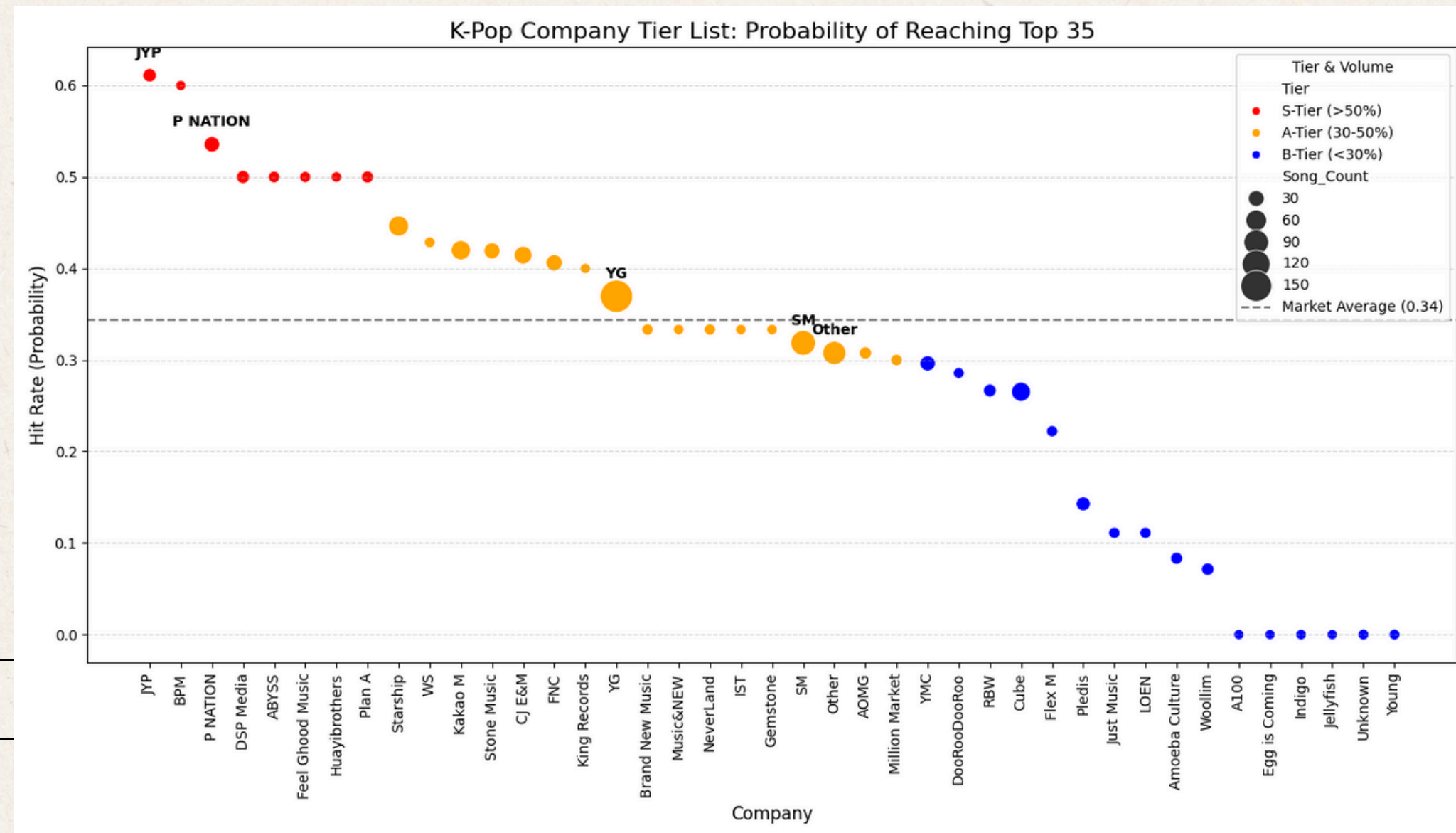
1. With only audio features, our regressors didn't work really well.
2. Also, the most of the important features are company names.

This suggests that in the K-Pop market, how the song sounds is not the primary determinant of its success.



Tier of Entertainment Companies

- JYP, SM, YG are seen as biggest entertainment companies in South Korea, however, songs from their artists are not always the hottest.
- P NATION, Starship, on the other hand, showing magnificent hit rate in second part of model training, although the scale of these two are not the highest.
- The phenomenon shows that big companies do not achieve success with every release.



Conclusion

Based on our data experiments and results, we found that music audio features alone are insufficient for predicting K-Pop success.

The key factors influencing a song's popularity extend beyond the track itself to include the artist marketing and fan factors, such as MV promotion and social media fan engagement.
